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标题	General and Efficient Copper-Catalyzed Oxazaborolidine Complex in Transfer Hydrogenation of Isoquinolines under Mild Conditions
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摘要（原文节选）

A general and efficient method for copper-catalyzed transfer hydrogenation of isoquinolines with an oxazaborolidine – BH3 complex, under mild reaction conditions, is successfully developed. A broad range of isoquinolines has been reduced to the corresponding products with 61 – 85% yields. The method is applied to the synthesis of biologically active tetrahydrosioquinoline alkaloid (±)-norlaudanosine in 62% yield, which is the key precursor for the preparation of (±)-laudanosine, (±)-N-methyl-laudanosine, and (±)-xylopi ⋯

数据表 · Table 1（22 行，含数值行 22 行）

Reaction Optimizationa

entry	metal	metal dosage (%)	hydrogenation source precursor	solvent	yield %b
1	CuCl	20	HSP1	THF	21
2	CuCl	20	HSP2	THF	14
3	CuCl	20	HSP3	THF	19
4	CuCl	20	HSP4	THF	24
5	CuCl	20	HSP5	THF	
6	CuCl	20	HSP6	THF	
7	CuCl	20	HSP7	THF	39
8	CuCl2	20	HSP7	THF	32
9	CuI	20	HSP7	THF	31
10	Cu(ClO4)2 · 6H2O	20	HSP7	THF	47
11	Cu(CF3SO3)2	20	HSP7	THF	79
12	Cu(OAc)2	20	HSP7	THF	38
13	CuSO4 · 5H2O	20	HSP7	THF	35
14	Cu(CF3SO3)2	15	HSP7	THF	81
15	Cu(CF3SO3)2	10	HSP7	THF	66
16	Cu(CF3SO3)2	5	HSP7	THF	55
17			HSP7	THF	21
18	Cu(CF3SO3)2	15	HSP7	DCM	61
19	Cu(CF3SO3)2	15	HSP7	DCE	85(80)
20	Cu(CF3SO3)2	15	HSP7	toluene	45
21	Cu(CF3SO3)2	15	HSP7	ether	32
22	Cu(CF3SO3)2	15	HSP7	dioxane	43

脚注（逐字）：aReaction conditions: substrate (0.5 mmol), boron reagents (2.0 equiv), solvent (2.0 mL); boron reagents prepared in situ: hydrogenation source precursor (2.0 equiv), BH3 · THF (4.0 equiv), 24 h, under Ar atmosphere.bYields were determined by GC using n-hexadecane as the internal calibration standard.

正文对应句（逐字）：Following our previous work, boron reagents was chosen as hydrogen source and isoquinoline 1a was chosen as model substrate. Boron reagents were generated in situ from BH3 · THF with various hydrogenation source precursor (Table 1, HSP1 – 7). As shown in Table 1, using cuprous chloride as catalyst and 1

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